

Down's Syndrome and impacts on memory and decision making

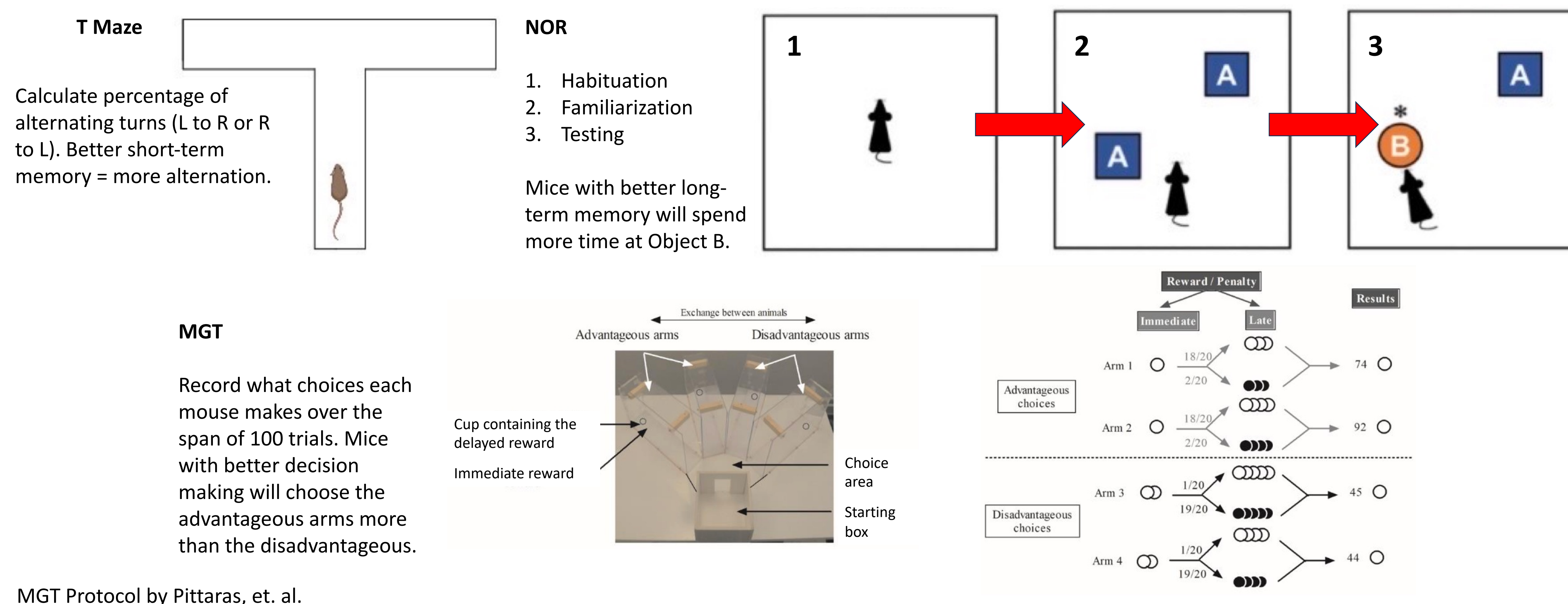
- Down's Syndrome, or Trisomy 21, is a common genetic disorder in which a person has a partial or complete extra copy of the 21st chromosome.
- Down's Syndrome is a complex disease affecting many systems, including:
 - Cardiovascular
 - Immune system
 - Cognition
 - Learning and memory
- Neurological impacts of Down's syndrome include negative impacts on both short- and long-term memory and decision-making.
- Ts65Dn mice have a trisomic chromosome orthologous to the 21st in humans.
- Ts65Dn mice have similar neurological symptoms.

Pentelynetetrazol: objectives

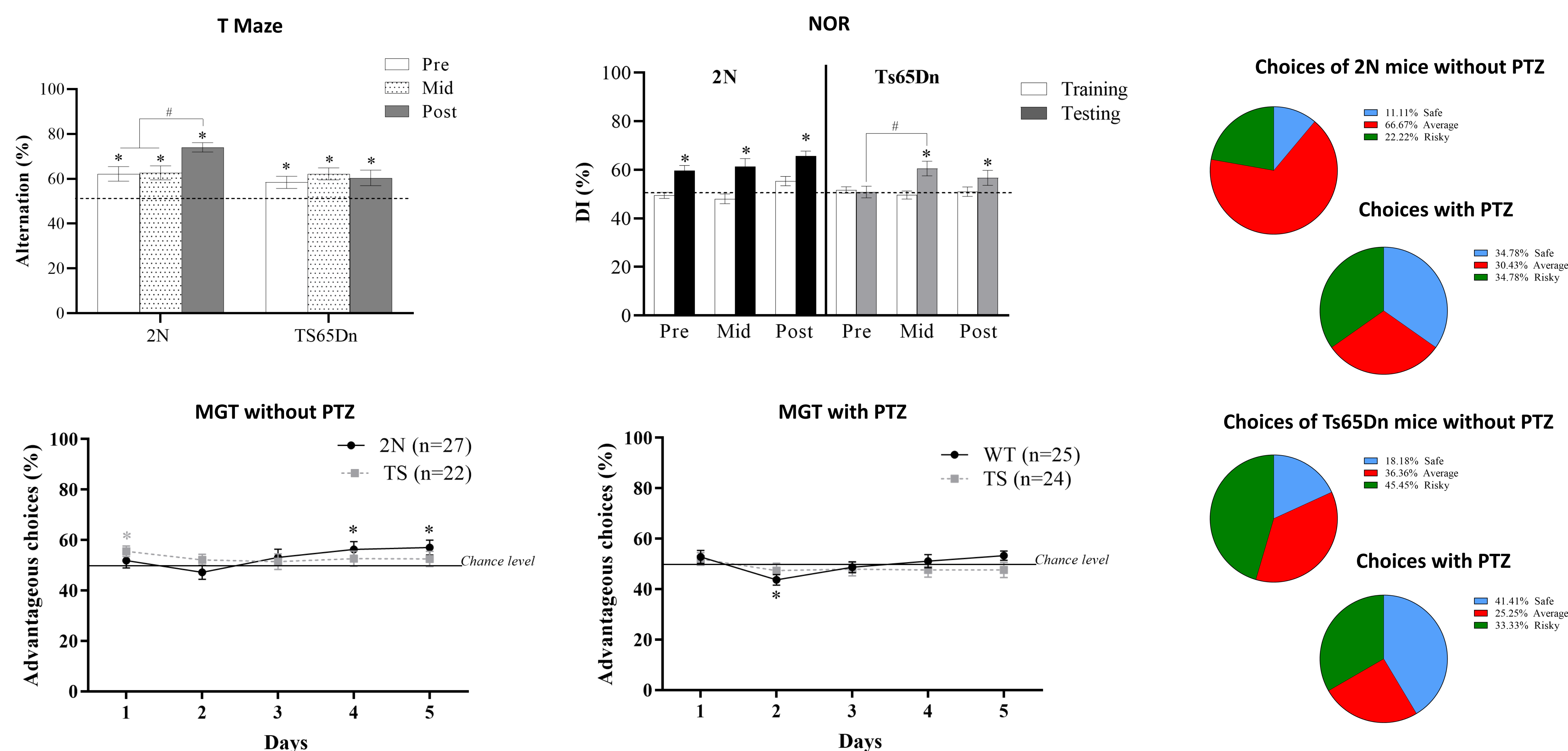
- Pentelynetetrazol (PTZ) is a GABA antagonist.
- Prior studies show that GABA antagonists improve short-term memory in Ts65Dn mice.
- We asked if dosing our cohort of Ts65Dn mice with PTZ would improve their long-term memory and decision-making skills over time in addition to short-term memory.**

Methods for testing memory and decision making

- T-Maze to test short-term memory
- Novel Object Recognition (NOR) to test long-term memory
- Mouse Gambling Task (MGT) to test decision making
- The mice were dosed with 3mg/kg of PTZ daily for 2 weeks, T-Maze, NOR, and MGT were performed before and after administering the PTZ. T-Maze and NOR were also performed in the middle of PTZ treatment.



Results



Conclusions

- The Ts65Dn mice showed statistically significant increases in NOR performance while they were being treated with PTZ, but after treatment ended, the significance of improvement went away.
- PTZ seems to decrease the decision-making performance for 2N mice during MGT.
- There was no significant change to decision making in Ts65Dn mice.
- Both the 2N and Ts65Dn cohorts made safer decisions during MGT after being treated with PTZ.

References

Pittaras E., Rabat A., Granon S. The Mouse Gambling Task: Assessing Individual Decision-making Strategies in Mice. Bio Protoc. 2020 Jan 5;10(1):e3479. doi: 10.21769/BioProtoc.3479. PMID: 33654712; PMCID: PMC7842597.

Acknowledgements

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